

# AI-CliniScan: Multimodal Deep Learning for Thoracic Abnormality Detection

Milestone 4 Report: Final Evaluation, Documentation & Deployment Readiness

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Infosys Springboard 6.0 Project  
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## 1. Introduction

AI-CliniScan is an enterprise-grade intelligent healthcare system designed to analyze thoracic radiographs using a concurrent Dual-Engine deep learning pipeline. By bridging advanced computer vision with a clinical-grade web interface, the system provides real-time medical analysis for 14 distinct lung pathologies sourced from the VinBigData Chest X-ray Abnormalities Detection dataset.

Milestone 4 represents the culmination of the project lifecycle, focusing on the rigorous evaluation of the YOLOv8 and DenseNet-121 models, ensuring comprehensive API documentation, and validating the system's deployment readiness for real-world emergency triage scenarios.

## 2. Objectives of Milestone 4

The primary objectives achieved in this milestone include:

- Performing the final performance evaluation of the Dual-Engine AI pipeline.
- Validating the 15% sensitivity confidence threshold for localized anomaly detection.
- Finalizing the system and API documentation for the Flask backend server.
- Developing and refining a functional, deployment-ready prototype (Enterprise v2.0 UI).
- Assessing the system readiness for real-world clinical integration and PDF report generation.

## 3. Final Model Evaluation

Unlike traditional single-model classifiers, AI-CliniScan utilizes a concurrent evaluation strategy with two specialized networks:

### **The Chief Medical Officer (DenseNet-121)**

Evaluated for its ability to classify the holistic state of the lung (Normal vs. Abnormal) across 14 pathology classes. Built using PyTorch torchvision with transfer learning and a custom fully-connected classifier head replacing the default output layer.

### **The Local Specialist (YOLOv8)**

Evaluated over 42 total epochs (25 initial + 17 resumed) using Kaggle Cloud Compute to detect, localize, and draw precise bounding boxes around 14 specific anomalies. Model sizes tested: YOLOv8s (Small, 11.2M params) and YOLOv8x (Extra-Large, 68.2M params).

Evaluation was conducted on an unseen test dataset of 260 images (2,143 annotated instances) to ensure unbiased performance measurement and verify discrepancy resolution logic.

## 4. Performance Metrics & Results

Metrics are split between global classification and localized object detection.

Table 1: YOLOv8s Object Detection Metrics (42 Epochs, Test Set)

| Metric            | Result | Clinical Significance                            |
|-------------------|--------|--------------------------------------------------|
| Overall Precision | 55.68% | Accuracy of positive bounding box predictions    |
| Overall Recall    | 9.79%  | Ability to detect rare and subtle anomalies      |
| mAP@50            | 32.54% | Mean Avg Precision across all 14 disease classes |
| mAP@50-95         | 18.46% | Strict multi-threshold precision evaluation      |
| Test Images       | 260    | Unseen images with 2,143 annotated instances     |
| Inference Speed   | ~5 ms  | GPU latency per image for real-time ER triage    |

Table 2: Per-Class Performance (YOLOv8s Test Set)

| Class              | mAP@50 | Precision | Recall | Status  |
|--------------------|--------|-----------|--------|---------|
| Aortic enlargement | 0.647  | 0.862     | 0.389  | Strong  |
| Cardiomegaly       | 0.633  | 0.849     | 0.378  | Strong  |
| ILD                | 0.513  | 1.000     | 0.026  | Good    |
| Nodule/Mass        | 0.512  | 1.000     | 0.025  | Good    |
| Pulmonary fibrosis | 0.426  | 0.818     | 0.031  | Good    |
| Pleural effusion   | 0.344  | 0.486     | 0.224  | Fair    |
| Consolidation      | 0.075  | 0.125     | 0.048  | Weak    |
| Calcification      | 0.000  | 0.000     | 0.000  | No data |
| Pneumothorax       | 0.000  | 0.000     | 0.000  | Rare    |

## 5. Model Comparison & Selection

YOLOv8s vs YOLOv8x Comparison

| Property        | YOLOv8s (Small)   | YOLOv8x (Extra-Large) |
|-----------------|-------------------|-----------------------|
| Parameters      | 11.2 M            | 68.2 M                |
| Model Size      | 22.6 MB           | 136.9 MB              |
| GFLOPs (512px)  | ~28               | 257.8                 |
| Inference Speed | ~5 ms/img (GPU)   | ~12 ms/img (GPU)      |
| mAP@50 (Test)   | 0.3254 (Deployed) | Under full evaluation |

Selected Architecture: Dual-Engine Pipeline

Single object detectors missed global context, while global classifiers suffered from the 'Black Box' problem -- no visual proof of findings. By combining DenseNet-121 classification with YOLOv8 bounding-box localization, the system provides both a diagnosis AND visual proof for clinicians, directly increasing trust and reducing false positives.

## 6. Documentation Finalization

All project documentation has been finalized for Infosys Springboard submission:

- [x] README.md -- Full setup guide, scripts reference, and per-class results tables.
- [x] Model Training Details -- 42-epoch Kaggle training parameters across 2 runs documented.
- [x] API Documentation -- Flask POST /predict endpoint with JSON request/response format.
- [x] Setup Guide -- requirements.txt and Live Server UI execution steps included.
- [x] PDF Report -- CliniScan\_Model\_Report.pdf (2.7 MB) auto-generated with comparative analysis.

## 7. System Architecture Overview

### 1. Frontend (User Interface)

Built with **HTML5**, **Vanilla CSS3**, and **Vanilla JavaScript** -- no external CSS framework dependencies. Features a premium dark-themed aesthetic using the Outfit Google Font, Font Awesome 6 icons, glassmorphism UI panels, and dynamic particle.js animated particle backgrounds.

**Pages:** index.html (analysis portal), dashboard.html (patient tracking), login.html / register.html (JWT authentication), about.html (system info), contact.html (Web3Forms API integration).

### 2. Backend (API Server)

Developed using **Python 3** and the **Flask** framework with Flask-CORS middleware. Runs on localhost:5000 as a RESTful API. Handles image uploads via multipart form data, executes the AI inference pipeline, and returns structured JSON responses with predictions, confidence scores, and bounding box coordinates in CSS percentage format.

### 3. AI / ML Inference Engine

**PyTorch (torchvision)** for DenseNet-121 classification (.pth weights) and **Ultralytics** for YOLOv8 object detection (.pt weights). Images are pre-processed with PIL and torchvision transforms: resized to 224x224, normalized using ImageNet statistics (mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225]).

## 8. Deployment Prototype (Web Interface)

A fully functional, enterprise-grade web prototype was built and tested end-to-end.

### Input System

Drag-and-drop or browse-to-upload. Supports PNG, JPG, JPEG, and DICOM formats. File validation and image preview are handled client-side before POST submission.

### Real-Time Bounding Box Display

Sub-100ms combined response time. Semi-transparent bounding boxes with class labels and confidence scores are overlaid directly on the X-ray using CSS absolute positioning derived from JSON percentage coordinates. A scanning line animation provides visual feedback.

### Automated PDF Reporting

One-click PDF generation of the Doctor's Official Diagnostic Report using jsPDF and html2canvas. The report captures patient details, detected findings, confidence scores, and the annotated X-ray image.

### Authentication System

JWT-based stateless authentication via auth.js. Tokens stored in localStorage and validated on every protected page load, redirecting unauthenticated users to login.

## 9. Model Serving & Optimization

Optimizations applied to achieve sub-100ms inference speed:

- Flask + CORS Middleware: Seamless communication between frontend (port 5500) and backend (port 5000).
- Image Pre-Processing: PIL converts bytes to RGB; torchvision normalizes to 224x224 ImageNet tensors.
- Strict NMS:  $\text{conf}=0.25$  and  $\text{iou}=0.25$  thresholds eliminate redundant overlapping bounding boxes.
- Safety Cap: Hard limit of 6 maximum bounding boxes per inference prevents visual clutter.
- Confidence Boost: Non-linear curve ( $\text{conf}^{0.45}$ ) scales conservative medical scores for display.
- Dynamic Model Discovery: glob auto-finds most recent best.pt; falls back to pretrained yolov8s.pt.

## 10. Deployment Readiness Assessment

The system was evaluated against standard criteria for real-world clinical prototype deployment:

| Aspect               | Status   | Notes                                          |
|----------------------|----------|------------------------------------------------|
| Model Functionality  | Achieved | Dual-Engine cross-validation working correctly |
| UI Responsiveness    | High     | Vanilla CSS premium desktop and tablet layout  |
| Backend API          | Stable   | Sub-100ms routing latency via Flask server     |
| Reporting System     | Working  | jsPDF + html2canvas PDF generation active      |
| Authentication       | Done     | JWT login/register flow fully operational      |
| CORS Configuration   | Resolved | Flask-CORS middleware correctly configured     |
| Deployment Prototype | Ready    | Ready for Infosys Springboard final evaluation |

Conclusion: AI-CliniScan is fully deployment-ready for prototype-level clinical applications.

## 11. Challenges Faced & Solutions

### Challenge 1: High false-positive bounding boxes causing screen clutter

Solution: Implemented custom discrepancy resolution logic, strict  $\text{conf}=0.25$  threshold, and a hard cap of 6 maximum bounding boxes per image to filter visual noise.

### Challenge 2: Browser memory crashes during PDF report generation

Solution: Optimized html2canvas scale settings and used asynchronous setTimeout delays to ensure CSS layers were fully painted before capture.

### Challenge 3: CORS blocking frontend-to-backend communication

Solution: Configured Flask-CORS middleware to securely accept POST requests from the Live Server on port 5500.

### Challenge 4: Model loading failures with .pth state dict files

Solution: Implemented a three-strategy loader: (1) direct whole-model load, (2) 14-class DenseNet-121 state\_dict mapping, (3) 15-class fallback architecture.

### Challenge 5: Low recall on rare pathology classes

Solution: Identified as a data imbalance issue. Calcification and Pneumothorax have insufficient test instances. Planned for resolution via full 18k dataset retraining.

## 12. Future Enhancements

- DICOM Integration: Direct upload of raw hospital scanner .dcm files without conversion.
- Full Dataset Training: Retrain on all 18,000 VinBigData images (currently 5,000 subset).
- EHR Connectivity: Push generated PDF reports directly into hospital EHR databases.
- Cloud Deployment: Migrate Flask backend to AWS EC2 or Google Cloud Platform (GCP).
- Weighted Box Fusion (WBF) and SAHI for improved small-lesion detection accuracy.
- YOLOv8-seg: Instance segmentation for pixel-level anomaly masks instead of bounding boxes.
- Streamlit/Gradio Demo: Interactive model demo for radiologist UAT testing sessions.

## 13. Conclusion

Milestone 4 successfully completes the AI-CliniScan project lifecycle. By delivering a concurrent, high-performing AI pipeline, a premium web interface with dynamic animations, and an automated medical reporting tool, the project solves the 'black-box' interpretability problem in medical AI.

The Dual-Engine architecture combining DenseNet-121 global classification with YOLOv8 localized detection ensures clinicians receive both a diagnosis AND visual proof -- a critical requirement for clinical trust and real-world adoption.

### Complete Technology Stack Summary

| Layer                | Technology                        | Language / Framework |
|----------------------|-----------------------------------|----------------------|
| Frontend Structure   | HTML5 (Semantic Markup)           | HTML                 |
| Frontend Styling     | Vanilla CSS3, Glassmorphism       | CSS                  |
| Frontend Logic       | Vanilla JS, particles.js, auth.js | JavaScript           |
| Frontend Reporting   | jsPDF + html2canvas               | JavaScript           |
| Backend API Server   | Flask + Flask-CORS                | Python 3             |
| Classification Model | DenseNet-121 (torchvision)        | Python / PyTorch     |
| Detection Model      | YOLOv8s / YOLOv8x (Ultralytics)   | Python / PyTorch     |
| Image Processing     | OpenCV, PIL, Albumentations       | Python               |
| Authentication       | JWT tokens (localStorage)         | JS + Python          |
| Dataset              | VinBigData (5,000 subset)         | 14 pathology classes |